

## ECE 313: Problem Set 3: Problems and Solutions

**Due:** Friday, February 21 at 07:00:00 p.m.

**Reading:** *ECE 313 Course Notes*, Sections 2.4 - 2.7.

**Note on reading:** For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

**Note on turning in homework:** Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:

NAME

NETID

SECTION

PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned pages.**

1. **[Elvis Had a Twin Brother]**

Elvis Presley had a twin brother who died at birth. This brother was either fraternal or identical. Using the fact that  $\frac{1}{125}$  births are fraternal twins and  $\frac{1}{300}$  are identical twins, find the probability that Elvis and his brother were identical twins, given that they were twins. Assume that boys and girls are equally likely for all births.

**Solution:** We know that Elvis had a twin brother. Since Elvis was also a man, that means that we need to consider the probability of a birth involving two identical twin brothers, compared to a birth where two brothers are twins in general. Let  $B$  be the event that two twins that are born are brothers, and let  $F$  and  $I$  denote the events of two twins being born that are fraternal or identical, respectively.

We can model the probability we need to find as

$$P(B \cap I | (B \cap I) \cup (B \cap F)) = \frac{P(B \cap I)}{P(B \cap I) + P(B \cap F)}$$

We can find these joint probabilities using the definition of conditional probability.

$$\begin{aligned} P(B \cap I) &= P(B|I)P(I) \\ &= \frac{1}{2} \frac{1}{300} \\ &= \frac{1}{600} \end{aligned}$$

Where  $P(B|I) = \frac{1}{2}$  because the only ways to have identical twins is for both to be brothers or both to be sisters, which are equally likely. Additionally,

$$\begin{aligned}
P(B \cap F) &= P(B|F)P(F) \\
&= \frac{1}{4} \cdot \frac{1}{125} \\
&= \frac{1}{500}
\end{aligned}$$

Where  $P(B|F) = \frac{1}{4}$  because fraternal twins have a quarter chance of being brothers, a quarter chance of being sisters, and a 50% chance of being siblings of opposite genders.

Putting this together, we find that

$$P(B \cap I | (B \cap I) \cup (B \cap F)) = \frac{\frac{1}{600}}{\frac{1}{600} + \frac{1}{500}} = \frac{5}{5+6} = \frac{5}{11}$$

**Alternative Solution/Intuition:** In a sample of 3000 births, for example, we would expect to see 24 fraternal twin pairs and 10 identical twin pairs.

The identical twins must be the same gender. There must be 5 brother pairs, and 5 sister pairs. The fraternal twins can be split into 6 brother pairs, 12 boy-girl pairs (in any order), and 6 sister pairs.

We are given that Elvis had a twin brother, meaning that he and his brother were either a brother fraternal pair or a brother identical pair. By counting, we can see that there are 11 twin brother pairs in our population, and 5 of those twin pairs are identical. Therefore, there is a  $\frac{5}{11}$  chance for Elvis to be in one of these identical twin brother pairs given that he was in one of these twin brother pairs.

Since all numbers of persons/pairs are proportional to 3000, the result is always  $\frac{5}{11}$  regardless of the sample size chosen, i.e. the required probability is  $\frac{5}{11}$ .

## 2. [Bernoulli RVs and Conditional Probability]

Let  $Z_1, Z_2, Z_3$  be independent Bernoulli random variables with parameters  $\frac{1}{3}, \frac{1}{3}, \frac{1}{2}$ , respectively. Define a random variable  $S = Z_1 + Z_2$  if  $Z_3 = 1$ , and  $S = 2Z_1 - Z_2$  if  $Z_3 = 0$ .

(a) Find  $P(S = 1)$ .

**Solution:** If  $Z_3 = 1$ , then  $S = 1$  when exclusively  $Z_1$  or  $Z_2$  is 1. If  $Z_3 = 0$ ,  $S = 1$  when  $Z_1 = Z_2 = 1$ .

$$\begin{aligned}
P(S = 1) &= P(Z_1 = 1, Z_2 = 0, Z_3 = 1) + P(Z_1 = 0, Z_2 = 1, Z_3 = 1) \\
&\quad + P(Z_1 = 1, Z_2 = 1, Z_3 = 0) \\
&= P(Z_1 = 1)P(Z_2 = 0)P(Z_3 = 1) + P(Z_1 = 0)P(Z_2 = 1)P(Z_3 = 1) \\
&\quad + P(Z_1 = 1)P(Z_2 = 1)P(Z_3 = 0) \\
&= \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{2} \\
&= \frac{5}{18}
\end{aligned}$$

(b) Find  $P(Z_3 = 1 | S = 1)$ .

**Solution:**

$$\begin{aligned}
 P(Z_3 = 1 | S = 1) &= \frac{P(Z_3 = 1, S = 1)}{P(S = 1)} \\
 &= \frac{P(Z_1 = 1, Z_2 = 0, Z_3 = 1) + P(Z_1 = 0, Z_2 = 1, Z_3 = 1)}{P(S = 1)} \\
 &= \frac{\frac{1}{3} \cdot \frac{2}{3} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{2}}{\frac{5}{18}} \\
 &= \frac{4}{5}.
 \end{aligned}$$

### 3. [Mathematicians and Matchbooks]

A pipe-smoking mathematician carries, at all times, 2 matchboxes, one in his left and another in his right pocket. Each time he needs a match to light the pipe he is equally likely to reach out to either of the two pockets. Both matchboxes start with  $N$  matches. Consider the moment when the mathematician discovers that one of his matchboxes is empty. What is the probability that there are exactly  $k$  matches in the other box, where  $k = 0, 1, \dots, N$ ?

**Solution:** This problem is more famously known as the “Banach Match Problem.”

The situation is the same regardless of whether the left or right matchbook is empty, so we can start by analyzing the case that the mathematician discovers that the right matchbook is empty and the left matchbook has  $k$  matches. Let’s call this event  $E$ .

In total, there are  $2N$  matches that the mathematician can light, and it’s a coinflip regarding whether he chooses the left or the right matchbox. If we consider choosing the right matchbox to be a “success” and the left matchbox to be a “failure,” we need to consider the probability that, after using  $N$  matches from the right matchbox and  $N - k$  matches from the left matchbox (in total  $2N - k$  matches), he reaches for the right match box on his  $2N - k + 1$ -th trial. With this construction, we can describe this situation as following a negative binomial distribution with  $2N - k + 1$  trials and  $N + 1$  “successes” with probability of success  $\frac{1}{2}$ .

Carefully note why it is  $N + 1$  successes and not  $N$ —even though on the last trial he wasn’t able to get a match, he still tried to use his right matchbook, which counts as a success under our model.

Then it’s simple to plug the correct terms into the distribution.

$$\begin{aligned}
 P(E) &= \binom{2N - k}{N} \left(\frac{1}{2}\right)^{N+1} \left(\frac{1}{2}\right)^{N-k} \\
 &= \binom{2N - k}{N} \left(\frac{1}{2}\right)^{2N-k+1}
 \end{aligned}$$

Because we need to evaluate both when the left and right matchbooks have  $k$ , and because this situation is symmetrical, we can double this value to get the probability of our final event.

$$\binom{2N - k}{N} \left(\frac{1}{2}\right)^{2N-k}$$

### 4. [First Success]

An experiment is conducted until it results in success: the first step has probability  $\frac{1}{2}$  to be

successful, the second step (only conducted if the first step was unsuccessful) has probability  $\frac{1}{3}$  to be successful, the third step (only conducted if the first two steps were unsuccessful) has probability  $\frac{1}{4}$  to be successful. If none of the steps were successful, we repeat the experiment until success is achieved. Assuming the first step has cost 2, and the second and third steps have cost 1, what is the expectation of the cost until success?

**Solution:** Let  $X$  denote the cost until success. We may find  $E[X]$  in a similar way as finding the expectation of the geometric distribution, by conditioning on the results of the first experiment (you may get the same expression using the law of total expectation, i.e. the formula at the beginning of page 58 in the textbook):

$$E[X] = 2P(\text{succeed at 1st step}) + 3P(\text{succeed at 2nd step}) \\ + 4P(\text{succeed at 3rd step}) + (E[4 + \tilde{X}])P(\text{fail in the first experiment})$$

where  $\tilde{X}$  denotes the additional cost if the first experiment fails and thus it has the same distribution as  $X$ . Therefore,

$$E[X] = 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} \cdot \frac{1}{3} + 4 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + (4 + E[X]) \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \\ = \frac{17}{6} + \frac{1}{4}E[X] \\ \implies E[X] = \frac{17}{6} \cdot \frac{4}{3} = \frac{34}{9}$$

## 5. [Geometric RVs]

Suppose your cousin owns a lightbulb manufacturing company and determines that 3 out of every 75 bulbs are defective in average. Your cousin now repeatedly conducts such a trial: select a lightbulb randomly from a great many lightbulbs and test if it is defective.

- (a) What is the probability that your cousin will find the first defective lightbulb on the 6th trial?

**Solution:** Let  $T$  denote the number of trials until the first defective lightbulb is found. We then know  $T$  follows the geometric distribution with parameter  $p = \frac{3}{75} = \frac{1}{25}$ .

$$P(T = 6) = (1 - p)^5 p = \frac{24^5}{25^6} \approx 0.0326$$

- (b) Now, what if your cousin wants to know the probability (likelihood) that it takes at least 6 trials until he/she/they find the first defective lightbulb?

**Solution:**

$$P(T \geq 6) = P(T > 5) = (1 - p)^5 = \frac{24^5}{25^5} \approx 0.815$$

- (c) Determine the expected number of trials your cousin has to conduct until he/she/they find the first defective, as well as the standard deviation.

**Solution:**

$$E[T] = \frac{1}{p} = \frac{1}{1/25} = 25 \\ \sigma_T = \frac{\sqrt{1-p}}{p} = \frac{(2\sqrt{6})/5}{1/25} = 10\sqrt{6} \approx 24.495$$

6. [Bonus Problem: Communication Systems and Binomial Distributions]

A communication system consists of  $n$  components, each of which independently functions with probability  $p$ . The total system is functional if at least half of the components are functional. For what values of  $p$  is a  $(2k + 1)$ -component system better than a  $(2k - 1)$ -component system? Your answer should be a number or numerical interval for  $p$ .

**Solution:** Consider a system of  $2k + 1$  components, and let  $X$  denote how many of the first  $2k - 1$  components function. We can understand 3 cases where the system with  $2k + 1$  components will be effective, based on  $X$ :

- $X \geq k + 1$
- $X = k$  and one of the two remaining components function.
- $X = k - 1$  and both of the two remaining components function

Then,

$$P_{2k+1}(\text{effective}) = P(X \geq k + 1) + P(X = k)(1 - (1 - p)^2) + P(X = k - 1)p^2$$

Let  $X$  have the same definition for the system of  $2k - 1$  components. Then,

$$P_{2k-1}(\text{effective}) = P(X \geq k) = P(X = k) + P(X \geq k + 1)$$

Finally, let's assume that  $P_{2k+1}(\text{effective})$  is greater than  $P_{2k-1}(\text{effective})$  and find the values of  $p$  that satisfy that assumption.

$$\begin{aligned} P_{2k-1}(\text{effective}) &< P_{2k+1}(\text{effective}) \\ 0 &< P_{2k+1}(\text{effective}) - P_{2k-1}(\text{effective}) \\ &= P(X = k)(1 - (1 - p)^2 - 1) + P(X = k - 1)p^2 \\ &= P(X = k - 1)p^2 - P(X = k)(1 - p)^2 \\ &= \binom{2k - 1}{k - 1} p^{k-1} (1 - p)^k p^2 - \binom{2k - 1}{k} p^k (1 - p)^{k-1} (1 - p)^2 \\ &= p \binom{2k - 1}{k} p^k (1 - p)^k - (1 - p) \binom{2k - 1}{k} p^k (1 - p)^k \\ &= (p - (1 - p)) \cdot \left( \binom{2k - 1}{k} p^k (1 - p)^k \right) \end{aligned}$$

The second factor is always positive, since  $0 < p < 1$ . Therefore this term is greater than zero as long as the first factor is positive, which is true when  $2p - 1$  is positive, i.e. when  $p > \frac{1}{2}$ .